

Northern Pomo searcher app: Accessing corpora through web-based applications

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Background - Northern Pomo is an indigenous language of Northern California. The last living speaker of Northern Pomo died in 2005. Despite this, there are still active efforts to document and revitalize the language. These efforts rest on a corpus of audio data recorded with the last living speakers by Eero Vihman (1960s) and Catherine O'Connor (1979 - 2005).

Digitizing the corpus – In total, the corpus contains ~ 900 hours of audio data. The original recordings were made on cassette tapes. A necessary first step was to convert the contents of these tapes to .wav files. This process is referred to as digitization. As each tape was digitized, the resulting file was uploaded to a Dropbox folder shared by a team of linguists.

Extraction and the Record of Tape
Extractions (RoTE) – After a sound file is digitized, we listen to it and identify individual Northern Pomo words and phrases. These are saved as separate sound files (a.k.a. extracted) to be displayed on a public [website](#). Both the source and extracted files are stored on Dropbox. To keep track of the source file for each word and phrase, the transcriptions and translations are entered into a Google Sheet titled **Record of Tape Extractions (RoTE)**.

Issues with the RoTE – At this time, the RoTE has over 10,000 entries. As such, it has become the single largest repository of text data for Northern Pomo. Additionally, its potential as a reference to the primary audio data makes it an invaluable resource. However, **the text data and the audio data are not directly linked**. This, coupled with idiosyncrasies of Google Sheets and Dropbox, make accessing the full corpus difficult.

Rote-searcher-app – In order to better integrate the text data in the RoTE and the audio data in Dropbox, I developed [rote-searcher-app](#). This is a python-based web application that allows the user to search the RoTE and receive output which includes the Northern Pomo transcription, a free translation in English, the source file that the utterance occurs in, a link to the source file in Dropbox, a link to a search for the extracted file in Dropbox, the time that the utterance occurred in the source file, any notes that were taken regarding the utterance at the time of extraction, and the row of the RoTE in which the entry appears. This process is sketched in the panel to the right.

Example search – Suppose you are looking for a word meaning “frayed”.

1 Enter “frayed” into the textbox labeled English and click **search**.

2 searches the English column in the RoTE Google Sheet for strings that match “frayed.”

ENGLISH	Northern Pomo	Speaker
*	*	EC
thread bare/ frayed	injini?	EC
thread bare/fraved	injini?	EC

Search the record of tape extractions

3 **Results:** matching entries are displayed in the browser window.

NPomo

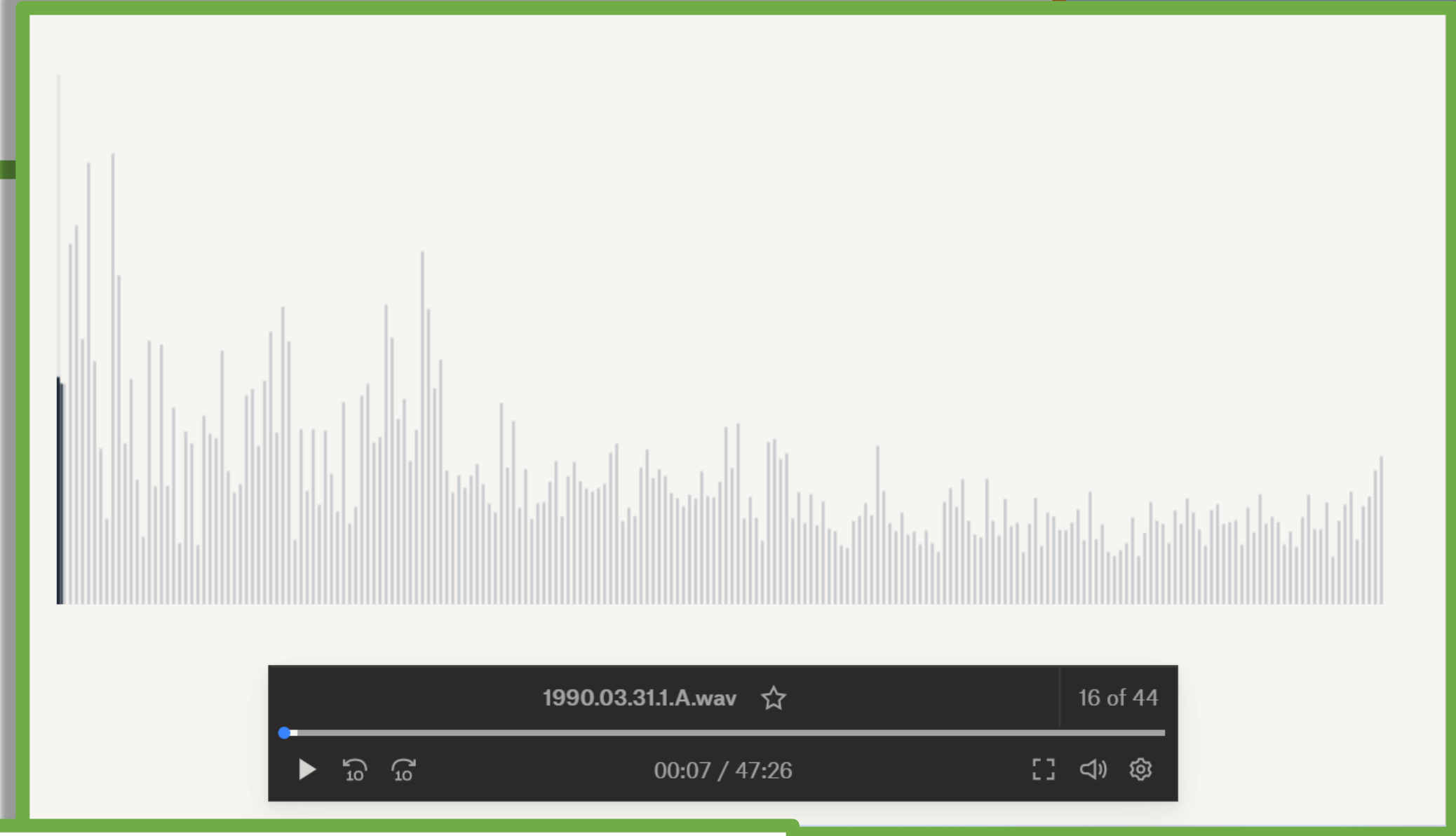
English

Speaker

Search

7 Results For: ('', 'frayed', '')

Npomo : injini?
Eng : thread bare/frayed
Speaker : EC
File : 1990.03.31.1.A
[Source File](#)
[Search Results](#)



4 Navigate to source file in Dropbox.

5 Navigate to search for the file containing extracted words and phrases in Dropbox.

thread bare frayed injini? EC.wav
words in isolation

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